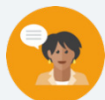


2.1 Square Values

Lesson Introduction

 Benchmark	MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.		 Learning Target	I can use multiple representations to explore square values.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real-number system.	
 Essential Questions	- What does it mean to square a number? - What is a perfect square? - How can perfect squares be represented geometrically? - How can perfect squares be represented algebraically?			
 Lesson Narrative	In this lesson, students make connections between geometric representations, algebraic equations, and verbal descriptions of perfect squares. Students draw squares on grid paper, represent the areas using $s^2 = A$, and connect this to the algebraic equation $x^2 = y$. Students substitute integers in this algebraic equation for x . Then they graph the coordinate pairs of the equation to discover that perfect squares are always positive. The conceptual foundation that the lesson lays should not be rushed. Allow students to explore and ask questions.			
 Component Summary	2.1.1 Warm-Up (~5 min.)	Reflect on growth mindset (<i>SE p. 63</i>)	2.1.3 Bring It Together (10-15 min.)	Determine what the term perfect square means (<i>SE p. 65</i>)
	2.1.2 Exploration (10 min.)	Relate the side lengths and areas of squares (<i>SE p. 64</i>)	2.1.4 Exploration (15 min.)	Explore the graph of $x^2 = y$ and values that satisfy this equation (<i>SE p. 66</i>)
	2.1.5 Wrap-Up (~5 min.)	Reflect on understanding and interest in lesson and write one lingering question (<i>SE p. 68</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms</u> : N/A <u>Additional Terms</u> : N/A		(Optional) Grid paper (Optional) Scissors	N/A

 Teacher Tips	Common Misconceptions - Students may multiply x by 2 instead of squaring it when substituting values in the equation $y = x^2$. - Students may plot coordinates as (y, x) instead of (x, y) when graphing $y = x^2$. - Students may not include the negative when squaring a negative value.	Things to Consider Students explore and create a definition for a perfect square. A perfect square is a number that results from multiplying a number by itself. It can also be called a square number. Consider building a common definition built around these ideas during the Stronger and Clearer Each Time routine in 2.1.3 Bring It Together.
	Literacy and Language Routines Stronger and Clearer Each Time 2.1.3 Bring It Together provides an opportunity for students to clarify and refine their definitions of perfect squares. After writing their initial definitions, students share them with a partner, who provides them with feedback. Students use the feedback to revise their definitions. This process is then repeated with different partners. Students' definitions and understandings improve as a result of the discussions and reflections that occur during this structured and interactive routine.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students first represent square values geometrically as the areas of squares in 2.1.2 Exploration. Then, in 2.1.4 Exploration, they represent square values using the algebraic equation $y = x^2$ and graph the (x, y) coordinate pairs. These multiple representations aid students in developing an understanding of perfect squares. Students also conclude that perfect squares are always positive. The progression from concrete to representational to abstract will promote student conceptual understanding and retention.
	 Differentiate Instruction	Students are provided with visual entry points in 2.1.2 and 2.1.4 Explorations as they graph the areas of squares and the equation $y = x^2$. Consider providing students with colored pencils and scissors so they can color and cut out their squares. This will provide a kinesthetic entry point. In 2.1.4 Exploration, allow students who struggle plotting points on the coordinate plane to use graphing technology, such as Desmos. Additional differentiation opportunities are denoted throughout the lesson guide.
Lesson Notes:		

2.1.1 Warm-Up: Reflecting on Learning

Component Overview: Students engage in a reflection focused on their development of a growth mindset.



2.1.1 Warm-Up: Reflecting on Learning

Learning something new can sometimes be challenging, but the challenges are part of what leads to success.

"Failure is not the opposite of success. It is part of success." -Arianna Huffington

1. What would you want your teacher to say to you when they see you taking a risk, trying hard, or pushing through mistakes to learn something new?

Answers will vary. I would want my teacher to encourage me to keep trying and recognize that I'm putting in my best effort even if I'm not there yet.

2. Describe how it feels when you learn or accomplish something that is really challenging.

Answers will vary. I feel proud of myself when I keep trying and accomplish something that was challenging to me. I feel like when I finally learn or accomplish something that was challenging that it was worth my time and effort.



Consider saving students' responses to question 1 and using their suggestions as positive reinforcements. Student responses to this question may provide insight into how they are motivated and encouraged.

Save students' responses to question 2 to motivate them when they are stuck on a challenging problem in the future.

2.1.2 Exploration: Area of Squares

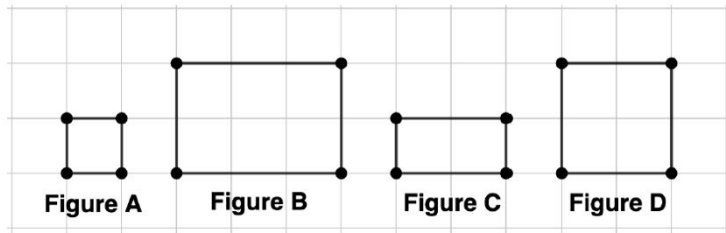
Component Overview: Students explore the relationship between the side lengths and areas of squares as they develop a sense of what it means for a number to be a square number.



2.1.2 Exploration: Area of Squares

Complete the following with your partner.

- Four figures are shown on the grid, where each gridline is 1 unit.



- Find the dimensions and area of each figure.

Figure	Dimensions	Area (units ²)
A	length=1 width=1	1
B	length=3 width=2	6
C	length=2 width=1	2
D	length=2 width=2	4

- All of the figures above are rectangles. Explain, using the dimensions, which of the shapes are also squares.

Figures A and D are also squares. Figure A has equal side measures of 1 unit. Figure D has equal side measures of 2 units.



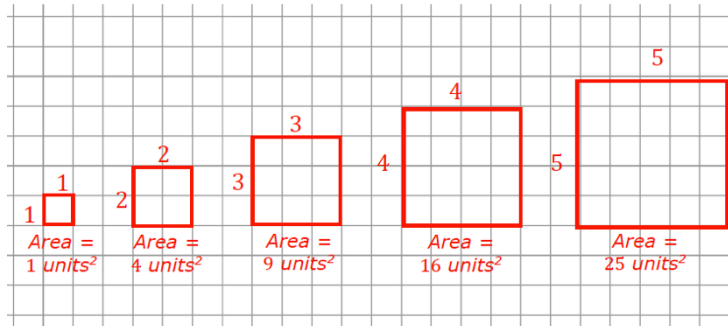
This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 2.1.3 Bring It Together.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

2.1.2 Exploration: Area of Squares (continued)

2. With your partner, create squares whose areas range from 1 to 25 square units. Create as many as possible on the grid paper below. Each gridline has a length of 1 unit.
 - a. For each square, label the length and the width, along with the area inside.



- b. Form a group with another pair of partners. Compare the squares each group created and, if necessary, add any squares to your grid.
3. Alicia says, "1 squared is 1 and 2 squared is 4." Explain what you think it means to square a number.
Answers will vary. I think it means to multiply a number by itself. For example, the square of 3 is 9 because it is the result of multiplying 3×3 .



To provide visual and kinesthetic entry points, consider allowing students to shade squares on the grid or to cut out squares with the given dimensions rather than drawing outlines.



It may be helpful to extend Alicia's definition by testing it with the other side lengths of the squares drawn in 2a. This may help prevent the common misconception of multiplying the side length by 2 instead of multiplying it by itself.

It is not the expectation that students write a perfect definition, rather that they are developing a sense of what it means for a number to be a square number. This concept will be further developed in the next component.

2.1.3 Bring It Together: Area of Squares

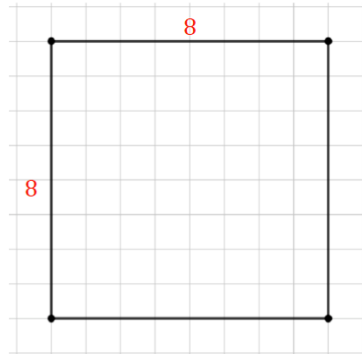
Component Overview: Students find the area of squares using $A = s^2$ and formalize their definition of what it means to be a perfect square. Students also begin to explore the idea that not all squares have side lengths that are whole numbers, a concept that is further developed in 2.1.4 Exploration.



2.1.3 Bring It Together: Area of Squares

1. The area of the square shown is 64 square units.

- a. Label the length and the width of the square on the grid.



- b. The length and width can be multiplied to find the area of 64 square units. Since the figure is a square, the area can also be found using $A = s^2$, where s is the length of a side.

Complete the equations for the square shown.

$$\text{Area} = \underline{(8)}^2 \quad \text{or} \quad \text{Area} = \underline{8} \times \underline{8}$$

2. Consider a square with an area of 100 units².

- a. What is the length of each side?

10 units

- b. Explain how you determined the length.

Answers will vary. We know that squares have equal side measures so the numbers we multiply to find the area has to be the same. $10 \times 10 = 100$, so each side has a measure of 10 units.



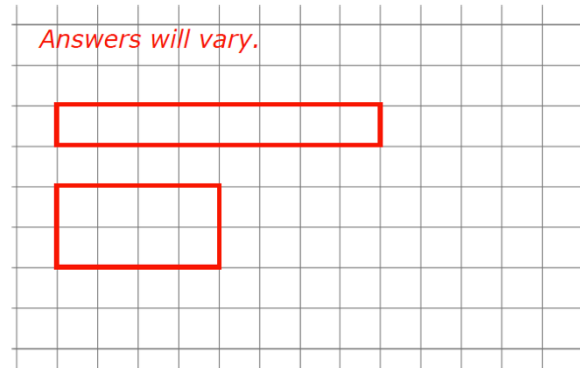
Students are not expected to take the square root of 100, as this skill will not be addressed until Lesson 2. Rather, prompt students to look for patterns between the side lengths and areas in 2.1.2 Exploration and their answers to question 1.

Consider asking questions such as:

- Looking back at the squares you drew in 2.1.2 Exploration, what relationship do you notice between the side lengths and areas?
- How could you use this relationship to determine the side length of the square an area of 100 units²?
- What number, when multiplied by itself, is equal to 100?

2.1.3 Bring It Together: Area of Squares (continued)

3. Consider a rectangle with an area of 8 square units.
- a. Use the grid to draw all of the possible rectangles with an area of 8 square units.



- b. Explain whether it is possible to have a square with an area of 8 square units.

Answers will vary. It is possible if the measure of the sides of the square are decimals or fractions. $2 \times 2 = 4$ and $3 \times 3 = 9$, a square with a side length that is somewhere between 2 and 3 could possibly make a square with an area of 8 square units.



Students are likely to consider only rectangles with whole-number side lengths (for example, 1×8 or 2×4). Highlight any students who happen to draw a square. The intent is not to introduce side lengths of $\sqrt{8}$. Rather, the idea is to prompt students to recognize that it is possible to have side lengths that are not whole numbers. If no students draw a square, consider prompting this realization by asking students the following after they respond to question 3b:

- *Are the side lengths of all quadrilaterals whole numbers?*
- *Are the side lengths of all squares whole numbers?*



Consider challenging students to draw a square with an area of 8 units². Remind students that they can partition the grid into fractional parts.

2.1.3 Bring It Together: Area of Squares (continued)

4. The numbers 1, 4, 16, 25, 64, and 100 are all examples of perfect squares. The numbers 3, 8, and 200 are nonexamples of perfect squares.
- a. Discuss with your partner what it means for a number to be a perfect square. Then, write your definition of a perfect square.
Answers will vary. A perfect square is a value found by squaring a whole number. For example, 25 is a perfect square because $5 \times 5 = 25$.
- b. Ask two other pairs of partners the definition they wrote. Listen to their definition and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of the Definition of a Perfect Square	Initials
<i>Answers will vary. A perfect square is the result of multiplying a whole number by itself.</i>	
<i>Answers will vary. A perfect square is a number that is made by multiplying by two of the same factors. For example, $3 \times 3 = 9$. 9 is a perfect square.</i>	

- c. Review the definitions with your partner and make any necessary adjustments to your own definition.



Stronger and Clearer Each Time

This routine provides an opportunity for students to clarify and refine their definitions of perfect squares. After writing their initial definitions, students share them with a partner, who provides them with feedback. Students use the feedback to revise their definitions. This process is then repeated with different partners. Students' definitions and understandings improve as a result of the discussions and reflections that occur during this structured and interactive routine.

Can you explain that in your own words?

- Is there something that is missing?*
- Can you explain this part more specifically?*
- How did you use the nonexamples to help write your definition?*

Also, encourage students to ask each other questions like this as they talk with each other to share definitions.

2.1.4 Exploration: Square Values on the Coordinate Plane

Component Overview: Students explore the equation and graph of $y = x^2$ to develop a deeper understanding of the relationship between a value and its square.



2.1.4 Exploration: Square Values on the Coordinate Plane

1. Use the equation $y = x^2$ to complete the following with your partner.
 - a. Complete the table of values for the equation $y = x^2$ by substituting the given values of x and solving for the corresponding y -value.

x	Substitute into $y = x^2$	Coordinates (x, y)
-3	$y = (-3)^2$ $y = 9$	$(-3, 9)$
-2	$y = (-2)^2$ $y = 4$	$(-2, 4)$
-1	$y = (-1)^2$ $y = 1$	$(-1, 1)$
0	$y = 0^2$ $y = 0$	$(0, 0)$
1	$y = 1^2$ $y = 1$	$(1, 1)$
2	$y = 2^2$ $y = 4$	$(2, 4)$
3	$y = 3^2$ $y = 9$	$(3, 9)$

- b. Discuss with your partner why the y -values are all perfect squares. Summarize your discussion.
Answers will vary. Each time we are multiplying a whole number by itself. Each time we are multiplying an integer by itself.



Consider discussing how $A = s^2$ and $y = x^2$ are alike. This may help students to connect the work done here with the geometric representations of square values that they've worked with up until this point.

The purpose of this activity is to expose students to the graph so they can build a deeper understanding of square numbers. This also extends from geometric squares, where the side length can only be positive values, to beginning to recognize that square values can be obtained by squaring either the positive or the negative square root. This concept will be further explored in the next lesson.

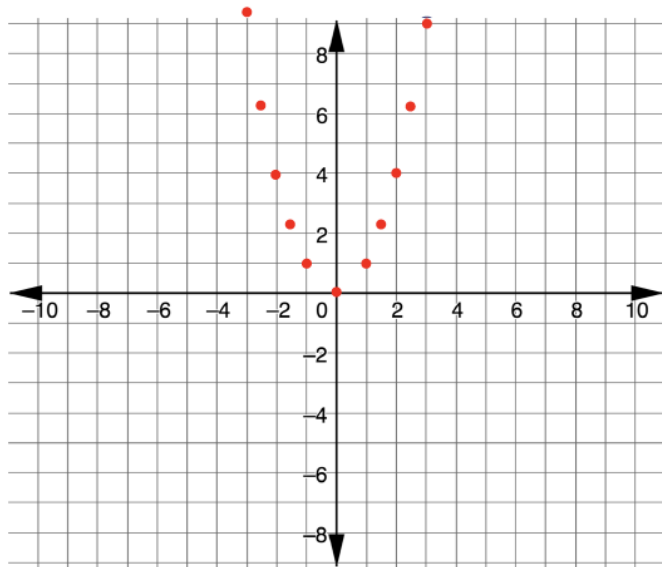


Consider asking the following to position students to notice patterns in the relationships:

- What do you notice about the y -values?
- What are some similarities and differences among all the coordinates?
- What is true about the x -values that correspond to the same y -values?

2.1.4 Exploration: Square Values on the Coordinate Plane (continued)

c. Plot the coordinates on the grid shown.



d. The graph of $y = x^2$ does not go below the x -axis. Complete the statements.

A positive number multiplied by itself results in a

- ☒ positive
- ☐ negative

product. A negative number

multiplied by itself results in a

- ☒ positive
- ☐ negative

product. Therefore, for any value of x multiplied by

itself, the value of y will

- ☐ sometimes
- ☒ always

be

- ☒ positive,
- ☐ negative,

so all of the points on the graph of

$y = x^2$ other than $(0,0)$ are above the x -axis.



Allow students who struggle plotting points on the coordinate plane to use a graphing utility.



Students are not expected to fully grasp what the graph of $y = x^2$ looks like or to describe it as a quadratic function or parabola. Rather, the goal is for students to see that squaring a positive or negative rational number always results in a positive rational number. Guide students to this conclusion with prompts such as:

- How does the graph help to prove that all the y -values in the table are positive?
- How does the graph show that the x -values that correspond to the same y -value are opposites?

2.1.4 Exploration: Square Values on the Coordinate Plane (continued)

2. Four coordinate pairs are shown. The points represented are also on the graph of $y = x^2$.

$$(1.5, 2.25) \quad (-1.5, 2.25) \quad \left(\frac{5}{2}, \frac{25}{4}\right) \quad \left(-\frac{5}{2}, \frac{25}{4}\right)$$

- a. Locate and plot the given points on the graph.
Shown on the graph.
- b. Discuss with your partner what you notice about the given coordinate pairs. Then, summarize your discussion.

Answers will vary. We noticed that the given coordinate pairs are all fractions or decimals, but the ordered pairs still line up with the curve of the graph. We noticed that they are still squaring the x-coordinate to get the y-coordinate and that opposite x-values still result in the same positive y-values once squared.

- c. Choose one of the points and show how it satisfies the equation $y = x^2$.

Answers will vary.

$$y = 1.5^2; 1.5 \times 1.5 = 2.25 \text{ or}$$

$$y = \left(-\frac{5}{2}\right)^2; y = -\frac{5}{2} \times -\frac{5}{2} = \frac{25}{4}$$



Students do not actually need to draw the parabola, as this goes beyond what they need to understand here. However, consider showing them the graph of $y = x^2$ on a graphing utility so that they understand the graph is curved.

2.1.5 Wrap-Up: Reflection

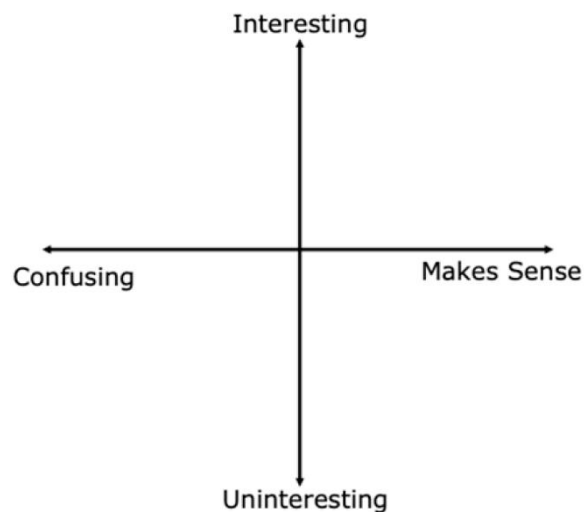
Component Overview: Students reflect on their understanding of and interest in the learning targets and identify a question they still have.



2.1.5 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and understanding of today's learning targets.

Answers will vary.



2. Write one question you have about square values after today's lesson.

Answers will vary. When we square a decimal or a fraction, are they called imperfect squares? If we need to find what makes the square of a number, should we put both the positive and negative number that will solve it?



Consider using the questions students write to form small groups for further instruction around concepts that students are still struggling with.